Work 3: Many-to-Many DRT with Bidirectional Meeting Points

Mathematical Model — Phase 1 Formulation

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Contents

1 Notation

Table 1: Summary of Notation

Symbol	Domain/Type	Description
Sets and Indices		
r	$\mathcal{R}$	Index for passenger requests; $\mathcal{R} = \{1, \dots, \lvert\mathcal{R}\rvert\}$
v	$\mathcal{V}$	Index for vehicles; $\mathcal{V} = \{1, \dots, \lvert\mathcal{V}\rvert\}$
$\mathcal{R}$	—	Set of all requests arriving in the planning horizon
$\mathcal{V}$	—	Set of all vehicles in the fleet
$\mathcal{M}$	—	Universe of pre-defined fixed meeting points (bus stops, landmarks)
M_r^P	$\subseteq \mathcal{M}$	Candidate pickup meeting point set for request r (filtered by walking ra
M_r^D	$\subseteq \mathcal{M}$	Candidate dropoff meeting point set for request r (filtered by walking ra
$\mathcal{B}_r$	—	Set of feasible service offer bundles for request r; $\mathcal{B}_r = M_r^P \times M_r^D$
Spatial and Temporal		
$d(i, j)$	$\mathbb{R}_{\geq 0}$	Euclidean distance between locations i and j
$t(i, j)$	$\mathbb{R}_{\geq 0}$	Travel time from i to j; $t(i, j) = d(i, j)/\bar{v}$ where $\bar{v}$ is average speed
o_r	—	Origin location of request r (passenger's pickup address)
δ_r	—	Destination location of request r (passenger's dropoff address)
e_r	$\mathbb{R}_{\geq 0}$	Earliest acceptable pickup time for request r (time window lower bound
l_r	$\mathbb{R}_{\geq 0}$	Latest acceptable pickup time for request r (time window upper bound)
ρ^P	$\mathbb{R}_{>0}$	Maximum walking radius for pickup meeting point (meters)
ρ^D	$\mathbb{R}_{>0}$	Maximum walking radius for dropoff meeting point (meters)
Service Offer Bundle		
b	$\mathcal{B}_r$	A feasible service offer bundle for request r
m_r^P	M_r^P	Selected pickup meeting point for request r
m_r^D	M_r^D	Selected dropoff meeting point for request r
τ_r	$\mathbb{R}_{\geq 0}$	Promised/scheduled pickup time at m_r^P
p_r	$\mathbb{R}_{\geq 0}$	Fare charged to request r (fixed per trip in this model)
Decision Variables		
z_r	$\{0, 1\}$	Acceptance indicator: $z_r = 1$ if request r is accepted, 0 otherwise
v_r	$\mathcal{V}$	Vehicle assigned to serve request r (defined when $z_r = 1$)
π_r^P	—	Insertion position of pickup node for r in vehicle v_r's route
π_r^D	—	Insertion position of dropoff node for r in vehicle v_r's route
Route and Schedule		
σ_v	—	Ordered route (sequence of nodes) for vehicle v
$s_{v,i}$	$\mathbb{R}_{\geq 0}$	Scheduled service time at node i in vehicle v's route
Q_v	$\mathbb{Z}_{>0}$	Passenger capacity of vehicle v
$q_v(i)$	$\mathbb{Z}_{\geq 0}$	Vehicle v's passenger load when departing from node i
$L^{\max}$	$\mathbb{R}_{>0}$	Maximum allowable in-vehicle ride time per passenger
$T_v^{\max}$	$\mathbb{R}_{>0}$	Maximum allowable route duration (shift length) for vehicle v

Continued on nex

Table 1 – continued

Symbol	Domain/Type	Description
Cost Components		
C^{op}	$\mathbb{R}_{\geq 0}$	Total vehicle operational cost (proportional to total distance driven)
C^{wait}	$\mathbb{R}_{\geq 0}$	Aggregate passenger waiting time cost
C^{walk}	$\mathbb{R}_{\geq 0}$	Aggregate passenger walking distance cost
C^{IVT}	$\mathbb{R}_{\geq 0}$	Aggregate in-vehicle travel time cost
C^{rej}	$\mathbb{R}_{\geq 0}$	Cost/penalty for each rejected request
$\alpha_1, \ldots, \alpha_5$	$\mathbb{R}_{\geq 0}$	Weights on cost components in objective function
Passenger Choice		
k	$\mathcal{K}$	Passenger type index; $\mathcal{K} = \{\text{price, time, walk}\}$
U_{rb}	$\mathbb{R}$	Deterministic utility of bundle b for request r
U_{r0}	$\mathbb{R}$	Utility of the outside option (reject all offers) for request r
P_{rb}	$[0, 1]$	MNL choice probability that request r selects bundle b
β_1^k	$\mathbb{R}$	Walk-time coefficient for passenger type k
β_2^k	$\mathbb{R}$	Waiting-time coefficient for passenger type k
β_3^k	$\mathbb{R}$	In-vehicle travel time (IVT) coefficient for passenger type k
β_4^k	$\mathbb{R}$	Price coefficient for passenger type k (negative for disutility)
Walk_{rb}	$\mathbb{R}_{\geq 0}$	Walking time to/from meeting points for request r under bundle b
Wait_{rb}	$\mathbb{R}_{\geq 0}$	Waiting time at pickup meeting point for request r under bundle b
IVT_{rb}	$\mathbb{R}_{\geq 0}$	In-vehicle travel time for request r under bundle b
Three-Layer Model		
H	$\mathbb{R}_{>0}$	Rolling horizon window length (minutes)
Δ	$\mathbb{R}_{>0}$	Rolling horizon time step (re-optimization frequency, minutes)
t^{now}	$\mathbb{R}_{\geq 0}$	Current system time in online operation
$\mathcal{R}^{\text{act}}(t)$	—	Set of active (accepted, not yet completed) requests at time t
k^{top}	$\mathbb{Z}_{>0}$	Number of top candidates generated per request per side (pickup and d

2 Problem Definition

Problem Definition

Network and Input

We model the service area as a directed graph $G = (N, A)$, where N is the set of nodes (passenger origins, destinations, and meeting points) and A is the set of directed arcs. Travel time between any two locations $i, j \in N$ is given by $t(i, j) = d(i, j)/\bar{v}$, where $d(i, j)$ denotes Euclidean distance and $\bar{v}$ is the average vehicle speed.

A finite set $\mathcal{M}$ of pre-defined fixed meeting points (e.g., bus stops, commercial landmarks) is given. For each request r, the candidate pickup set $M_r^P \subseteq \mathcal{M}$ is the subset of meeting points within walking radius ρ^P of the passenger's origin o_r, and the candidate dropoff set $M_r^D \subseteq \mathcal{M}$ is the subset within walking radius ρ^D of the destination δ_r. Formally:

$$M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}, \qquad M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}. \tag{1}$$

Each request $r \in \mathcal{R}$ is characterized by a tuple $(o_r, \delta_r, e_r, l_r, k_r)$, where $[e_r, l_r]$ is the pickup time window and $k_r \in \mathcal{K}$ denotes the passenger type. Requests arrive online over a continuous planning horizon.

Service Offer Bundle

For each request r, a *service offer bundle* is a tuple

$$b = (m_r^P,\ m_r^D,\ \tau_r,\ p_r),$$

where:

- $m_r^P \in M_r^P$ is the assigned pickup meeting point,
- $m_r^D \in M_r^D$ is the assigned dropoff meeting point,
- $\tau_r \in [e_r, l_r]$ is the scheduled pickup time at m_r^P,
- $p_r \geq 0$ is the fare charged to the passenger (fixed in this model).

The set of all feasible bundles for request r is $\mathcal{B}_r = M_r^P \times M_r^D$ (with τ_r and p_r treated as service parameters determined by the system). The service offer bundle b is the central decision object: the system selects one bundle from $\mathcal{B}_r$ or rejects the request.

System Objective

The system minimizes a weighted sum of operational and passenger service costs over the planning horizon:

$$\min \quad \alpha_1 C^{op} + \alpha_2 C^{wait} + \alpha_3 C^{walk} + \alpha_4 C^{IVT} + \alpha_5 C^{rej}, \tag{2}$$

where:

$$C^{op} = \sum_{v \in \mathcal{V}} c_v \cdot D_v, \tag{3}$$

$$C^{wait} = \sum_{r \in \mathcal{R}} z_r \cdot (\tau_r - t_r^{\text{req}}), \tag{4}$$

$$C^{walk} = \sum_{r \in \mathcal{R}} z_r \cdot \big(d(o_r, m_r^P) + d(m_r^D, \delta_r)\big), \tag{5}$$

$$C^{IVT} = \sum_{r \in \mathcal{R}} z_r \cdot \text{IVT}_{rb}, \tag{6}$$

$$C^{rej} = \sum_{r \in \mathcal{R}} (1 - z_r) \cdot \Gamma. \tag{7}$$

Here D_v is the total distance driven by vehicle v, c_v is cost per unit distance, t_r^{req} is the time request r was submitted, and $\Gamma > 0$ is the rejection penalty.

Online Decision Vector

Upon arrival of request r, the system must immediately determine the *online decision vector*:

$$\mathbf{x}_r = (z_r,\ m_r^P,\ m_r^D,\ v_r,\ \pi_r^P,\ \pi_r^D), \tag{8}$$

where:

- $z_r \in \{0,1\}$ is the acceptance indicator (1 = accepted, 0 = rejected),
- $m_r^P \in M_r^P$ is the pickup meeting point (defined when $z_r = 1$),
- $m_r^D \in M_r^D$ is the dropoff meeting point (defined when $z_r = 1$),
- $v_r \in \mathcal{V}$ is the vehicle assigned to serve request r (defined when $z_r = 1$),
- π_r^P is the insertion position of the pickup node in vehicle v_r's route,
- π_r^D is the insertion position of the dropoff node in vehicle v_r's route, with $\pi_r^D > \pi_r^P$ (pickup precedes dropoff in route).

The decision must be made online — i.e., before future requests are known — and subject to all feasibility constraints specified in Section 3.

3 Operational Constraints

Operational Constraints

The following constraints define the feasibility region for the many-to-many DRT problem with bidirectional meeting points. All constraints apply to accepted requests ($z_r = 1$) unless otherwise noted. Constraint labels are used directly in the Phase 2 MILP formulation and the fast feasibility checker.

Vehicle Capacity

At no point in any vehicle's route may the number of passengers on board exceed the vehicle's capacity.

$$q_v(i) \le Q_v \quad \forall v \in \mathcal{V},\ \forall i \in \sigma_v \tag{9}$$

where $q_v(i)$ is the passenger load on vehicle v when departing from node i, and Q_v is the vehicle's maximum capacity.

Time Windows

Each accepted request must be picked up within its specified time window at the assigned pickup meeting point.

$$s_{v_r, m_r^P} \ge e_r \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{10}$$

$$s_{v_r, m_r^P} \le l_r \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{11}$$

where s_{v_r, m_r^P} is the scheduled service time at pickup meeting point m_r^P of vehicle v_r, and $[e_r, l_r]$ is the pickup time window for request r.

In-Vehicle Ride Time

The time a passenger spends riding in the vehicle (from pickup meeting point to dropoff meeting point) must not exceed the maximum allowable ride time.

$$s_{v_r,\, m_r^D} - s_{v_r,\, m_r^P} \le L^{\max} \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{12}$$

where s_{v_r, m_r^D} is the scheduled service time at dropoff meeting point m_r^D, and $L^{\max}$ is the maximum allowable in-vehicle ride time.

Walking Radius

The assigned pickup and dropoff meeting points must lie within the maximum walking radius of the passenger's origin and destination, respectively.

$$d(o_r,\, m_r^P) \le \rho^P \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{13}$$

$$d(\delta_r,\, m_r^D) \le \rho^D \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{14}$$

where $d(o_r, m_r^P)$ is the Euclidean distance from origin o_r to pickup meeting point m_r^P, ρ^P is the maximum pickup walking radius, and ρ^D is the maximum dropoff walking radius.

Note: These constraints are equivalent to the candidate set filters M_r^P and M_r^D defined in the problem setup; they are stated here explicitly to make the MILP formulation complete.

Precedence: Pickup Before Dropoff

For each accepted request, the pickup meeting point node must appear before the dropoff meeting point node in the assigned vehicle's route, and the vehicle must serve the pickup before the dropoff in time.

$$\pi_r^P < \pi_r^D \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{15}$$

$$s_{v_r,\, m_r^P} + t(m_r^P,\, m_r^D) \le s_{v_r,\, m_r^D} \quad \forall r \in \mathcal{R},\ z_r = 1 \tag{16}$$

where π_r^P and π_r^D are the route insertion positions of the pickup and dropoff nodes (both on vehicle v_r), and $t(m_r^P, m_r^D)$ is the travel time from pickup to dropoff meeting point (a lower bound on the in-vehicle segment).

Route Time Consistency

Scheduled service times must be consistent with travel times along each vehicle's route — the vehicle cannot arrive at a node before it has had time to travel from the previous node.

$$s_{v,\,i+1} \ge s_{v,\,i} + t(\sigma_v(i),\, \sigma_v(i+1)) \quad \forall v \in \mathcal{V},\ \forall i < |\sigma_v| \tag{17}$$

where $\sigma_v(i)$ is the i-th node in vehicle v's route, and $t(\sigma_v(i), \sigma_v(i+1))$ is the travel time from the i-th to the $(i+1)$-th node.

Maximum Route Duration

The total operating time of each vehicle from its first service node to its last must not exceed the vehicle's maximum shift duration.

$$s_{v,\,|\sigma_v|} - s_{v,\,1} \leq T_v^{\max} \quad \forall v \in \mathcal{V} \tag{18}$$

where $s_{v,1}$ is the scheduled time at the first node in vehicle v's route, $s_{v,|\sigma_v|}$ is the scheduled time at the last node, and $T_v^{\max}$ is the maximum allowable route duration (shift length) for vehicle v.

Domain Constraints

The decision variables are restricted to the following domains.

$$z_r \in \{0,1\} \qquad \forall r \in \mathcal{R} \tag{19}$$

$$v_r \in \mathcal{V} \qquad \forall r \in \mathcal{R},\ z_r = 1 \tag{20}$$

$$m_r^P \in M_r^P \qquad \forall r \in \mathcal{R},\ z_r = 1 \tag{21}$$

$$m_r^D \in M_r^D \qquad \forall r \in \mathcal{R},\ z_r = 1 \tag{22}$$

Constraint Summary

Label	Class	Variables Involved
9	Vehicle Capacity	$q_v(i)$, Q_v, σ_v
10	Time Windows (early)	s_{v_r,m_r^P}, e_r
11	Time Windows (late)	s_{v_r,m_r^P}, l_r
12	In-Vehicle Ride Time	s_{v_r,m_r^D}, s_{v_r,m_r^P}, $L^{\max}$
13	Walking Radius (pickup)	$d(o_r, m_r^P)$, ρ^P
14	Walking Radius (dropoff)	$d(\delta_r, m_r^D)$, ρ^D
15	Precedence (position)	π_r^P, π_r^D
16	Precedence (time)	s_{v_r,m_r^P}, s_{v_r,m_r^D}, $t(m_r^P, m_r^D)$
17	Route Time Consistency	$s_{v,i}$, $s_{v,i+1}$, $t(\sigma_v(i), \sigma_v(i+1))$
18	Maximum Route Duration	$s_{v,1}$, $s_{v,|\sigma_v|}$, $T_v^{\max}$
19	Domain (acceptance)	z_r
20	Domain (vehicle)	v_r
21	Domain (pickup point)	m_r^P, M_r^P
22	Domain (dropoff point)	m_r^D, M_r^D

4 Passenger Choice Model

Passenger Choice Model

Utility Function

We model passenger choice using a Multinomial Logit (MNL) model. For request r of passenger type $k \in \mathcal{K}$, the deterministic utility of service offer bundle $b = (m_r^P, m_r^D, \tau_r, p_r)$ is:

$$U_{rb}^k = \beta_1^k \cdot \mathrm{Walk}_{rb} + \beta_2^k \cdot \mathrm{Wait}_{rb} + \beta_3^k \cdot \mathrm{IVT}_{rb} + \beta_4^k \cdot p_r \tag{23}$$

where:

- $\mathrm{Walk}_{rb} = d(o_r, m_r^P) + d(m_r^D, \delta_r)$ is the total walking distance (pickup walk + dropoff walk) in meters,
- $\mathrm{Wait}_{rb} = \tau_r - t_r^{\mathrm{req}}$ is the waiting time from request submission to scheduled pickup (minutes),
- $\mathrm{IVT}_{rb} = s_{v_r, m_r^D} - s_{v_r, m_r^P}$ is the in-vehicle travel time from pickup to dropoff meeting point (minutes),
- $p_r \geq 0$ is the fare (CNY),
- $\beta_1^k, \beta_2^k, \beta_3^k \leq 0$ are disutility coefficients for walking, waiting, and in-vehicle time respectively (negative: more time/distance = lower utility),
- $\beta_4^k \leq 0$ is the price disutility coefficient (negative: higher fare = lower utility).

The random utility is $U_{rb}^k + \varepsilon_{rb}$ where ε_{rb} is i.i.d. Gumbel$(0, 1)$, yielding the standard MNL form.

Outside Option

The choice set for request r includes an outside option representing the passenger's decision to reject all offered bundles (e.g., use a private car, walk, or cancel the trip). The utility of the outside option is:

$$U_{r0}^k = \beta_0^k \tag{24}$$

where β_0^k is the alternative-specific constant (ASC) for the outside option of passenger type k. Setting $\beta_0^k = 0$ normalizes the outside option as the reference alternative; a positive β_0^k indicates a preference for the outside option (e.g., passengers with private car access).

The full choice set for request r is $\mathcal{B}_r \cup \{0\}$, where 0 denotes the outside option.

Passenger Heterogeneity

We distinguish three passenger types $\mathcal{K} = \{\text{price}, \text{time}, \text{walk}\}$ with different sensitivity profiles. Table 2 summarizes the qualitative ordering of coefficients; exact values are calibrated from literature or simulation in Phase 3.

Table 2: Passenger type coefficient profiles (qualitative ordering)

Type	β_1^k (Walk)	β_2^k (Wait)	β_3^k (IVT)	β_4^k (Price)
Price-sensitive	moderate	moderate	moderate	high (most nega
Time-sensitive	moderate	high (most negative)	high (most negative)	low
Walk-sensitive	high (most negative)	moderate	moderate	moderate

Formally, the ordering constraints are:

$$|\beta_4^{\text{price}}| > |\beta_4^{\text{time}}|, \quad |\beta_4^{\text{price}}| > |\beta_4^{\text{walk}}| \tag{25}$$

$$|\beta_2^{\text{time}}| + |\beta_3^{\text{time}}| > |\beta_2^{\text{price}}| + |\beta_3^{\text{price}}| \tag{26}$$

$$|\beta_1^{\text{walk}}| > |\beta_1^{\text{price}}|, \quad |\beta_1^{\text{walk}}| > |\beta_1^{\text{time}}| \tag{27}$$

Baseline parameter values for simulation (provisional — to be confirmed against Work 1/2 calibration before Phase 3):

$$(\beta_1^{\text{price}}, \beta_2^{\text{price}}, \beta_3^{\text{price}}, \beta_4^{\text{price}}) = (-0.005,\ -0.04,\ -0.02,\ -0.15) \tag{28}$$

$$(\beta_1^{\text{time}}, \beta_2^{\text{time}}, \beta_3^{\text{time}}, \beta_4^{\text{time}}) = (-0.005,\ -0.10,\ -0.08,\ -0.03) \tag{29}$$

$$(\beta_1^{\text{walk}}, \beta_2^{\text{walk}}, \beta_3^{\text{walk}}, \beta_4^{\text{walk}}) = (-0.020,\ -0.04,\ -0.02,\ -0.05) \tag{30}$$

Units: β_1 in utility/meter, β_2 and β_3 in utility/minute, β_4 in utility/CNY.

Choice Probability and Acceptance Rate

Given the choice set $\mathcal{B}_r \cup \{0\}$ and utilities $\{U_{rb}^k\}_{b\in\mathcal{B}_r}$ and U_{r0}^k, the MNL choice probability that passenger r (of type k) selects bundle b is:

$$P_{rb}^k = \frac{\exp(U_{rb}^k)}{\exp(U_{r0}^k) + \sum_{b'\in\mathcal{B}_r} \exp(U_{rb'}^k)} \quad \forall b \in \mathcal{B}_r \tag{31}$$

The probability of choosing the outside option (rejecting all bundles) is:

$$P_{r0}^k = \frac{\exp(U_{r0}^k)}{\exp(U_{r0}^k) + \sum_{b'\in\mathcal{B}_r} \exp(U_{rb'}^k)} \tag{32}$$

The overall acceptance probability for request r is:

$$P_r^{\text{acc},k} = 1 - P_{r0}^k = \frac{\sum_{b\in\mathcal{B}_r} \exp(U_{rb}^k)}{\exp(U_{r0}^k) + \sum_{b'\in\mathcal{B}_r} \exp(U_{rb'}^k)} \tag{33}$$

In the online decision process, the system selects the bundle $b^* \in \mathcal{B}_r$ that maximizes expected utility (or expected system objective contribution), then presents it to the passenger. The passenger accepts with probability $P_{rb^*}^k$ and rejects with probability P_{r0}^k. The realized acceptance z_r is therefore a Bernoulli random variable with parameter $P_{rb^*}^k$.

5 Three-Layer Coupled Model

Three-Layer Coupled Model

The many-to-many DRT system with bidirectional meeting points operates through three sequentially coupled layers, executed online upon each request arrival and periodically re-optimized via a rolling horizon mechanism. Figure ?? (Phase 6) illustrates the architecture. The three layers are: (1) service offer generation, (2) passenger response, and (3) dynamic dispatch and re-optimization.

Layer 1: Service Offer Generation

Upon arrival of request r at time t_r^{req}, the system generates a set of candidate service offer bundles $\hat{\mathcal{B}}_r \subseteq \mathcal{B}_r$ as follows:

1. **Candidate pickup set:** Compute $M_r^P = \{m \in \mathcal{M} \mid d(o_r, m) \leq \rho^P\}$. Retain the top-$k^{\text{top}}$ candidates ranked by proximity to o_r.

2. **Candidate dropoff set:** Compute $M_r^D = \{m \in \mathcal{M} \mid d(\delta_r, m) \leq \rho^D\}$. Retain the top-$k^{\text{top}}$ candidates ranked by proximity to δ_r.

3. **Bundle enumeration:** For each pair $(m^P, m^D) \in M_r^P \times M_r^D$, evaluate feasibility against constraints 9–17 for each vehicle $v \in \mathcal{V}$ and each insertion position pair (π^P, π^D).

4. **Bundle selection:** Select the bundle $b^* = \arg\max_{b \in \hat{\mathcal{B}}_r} \mathbb{E}[\text{system benefit} \mid b, \mathbf{x}_r]$ that maximizes expected system objective contribution, subject to feasibility.

The output of Layer 1 is the online decision vector $\mathbf{x}_r = (z_r, m_r^P, m_r^D, v_r, \pi_r^P, \pi_r^D)$ (Equation 8) and the service offer bundle b^* presented to the passenger.

Layer 2: Passenger Response

Given the offered bundle b^*, passenger r of type k accepts with probability $P_{rb^*}^k$ (Equation 31) and rejects (chooses the outside option) with probability P_{r0}^k (Equation 32).

The realized acceptance indicator is:

$$z_r \sim \text{Bernoulli}(P_{rb^*}^k) \tag{34}$$

If $z_r = 1$, the vehicle route σ_{v_r} is updated by inserting the pickup node m_r^P at position π_r^P and the dropoff node m_r^D at position π_r^D. If $z_r = 0$, the decision vector is discarded and the vehicle routes remain unchanged.

The coupling between Layer 1 and Layer 2 is *anticipatory*: the system selects b^* to maximize expected system benefit, accounting for the probability that the passenger will accept. This distinguishes the model from a deterministic assignment approach.

Layer 3: Dynamic Dispatch and Rolling Horizon Re-optimization

Vehicle dispatch follows the committed route schedules $\{\sigma_v, \mathbf{s}_v\}_{v \in \mathcal{V}}$. Every Δ minutes, a rolling horizon re-optimization is triggered over the active request window $[t^{\text{now}},\ t^{\text{now}} + H]$:

1. **Active set:** Identify $\mathcal{R}^{\text{act}}(t^{\text{now}})$ — all accepted requests not yet completed at time t^{now}.

2. **Re-optimization:** Apply ALNS (Adaptive Large Neighborhood Search) to reassign and reorder requests in $\mathcal{R}^{\text{act}}(t^{\text{now}})$ across vehicles, subject to constraints 9–17.

3. **Commitment:** Commit the updated routes for the next Δ minutes; nodes already served are locked and cannot be reassigned.

The rolling horizon creates a feedback loop from Layer 3 back to Layer 1: the updated vehicle schedules after re-optimization change the feasibility and cost of future bundle insertions, influencing the next Layer 1 evaluation.

$$\text{Re-optimize at times } t^{\text{now}} = 0,\ \Delta,\ 2\Delta,\ 3\Delta,\ \ldots \tag{35}$$

Typical parameter values: $H = 30$ minutes (horizon window), $\Delta = 5$ minutes (re-optimization frequency). These are configurable and subject to sensitivity analysis in Phase 4.

Layer Coupling Summary

Table 3: Three-layer model coupling structure

Layer	Input	Output	Coupling
1: Service Generation	Request r, routes $\{\sigma_v\}$	Bundle b^*, decision $\mathbf{x}_r$	Feeds Layer 2
2: Passenger Response	Bundle b^*, type k	Acceptance z_r	Updates routes if $z_r = 1$
3: Rolling Horizon	Active set $\mathcal{R}^{\text{act}}$	Updated routes $\{\sigma_v\}$	Feeds back to Layer 1

6 Summary and Phase 2 Handoff

This document constitutes the complete mathematical foundation for Work 3. Table 4 maps each model component to the corresponding Phase 2 implementation requirement.

Table 4: Phase 1 to Phase 2 handoff map

Model Component	Section	Phase 2 Requirement
Notation table	§1	All (symbol reference)
Network + meeting point sets	§2	HEUR-01 (candidate generation)
Service offer bundle	§2	EXACT-01, HEUR-02
System objective	§2	EXACT-01 (MILP objective)
Online decision vector	§2	HEUR-02 (insertion evaluator)
Capacity constraint	§3	EXACT-01, HEUR-03
Time window constraints	§3	EXACT-01, HEUR-03
Ride time constraint	§3	EXACT-01, HEUR-03
Walking radius constraints	§3	HEUR-01 (filter)
Precedence constraints	§3	EXACT-01, HEUR-03
Route time consistency	§3	EXACT-01, HEUR-03
Maximum route duration	§3	EXACT-01, HEUR-03
MNL utility function	§4	Phase 3 simulation
Passenger type profiles	§4	Phase 3 simulation
Choice probability	§4	HEUR-02 (expected benefit)
Layer 1: Service generation	§5	HEUR-01, HEUR-02
Layer 2: Passenger response	§5	HEUR-02
Layer 3: Rolling horizon	§5	HEUR-04, HEUR-05